

MAT 534 HW07

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Due 10/29 (extended to 10/31)

Problem 1. Let F be a field. Prove that the polynomial ring $F[x]$ is a PID.

Solution. Let $I \subseteq F[x]$ be an ideal and $f(x) \in I$ be a polynomial whose degree is as small as possible. Then for any $g(x) \in I$, we can divide g by f to get $g(x) = f(x)q(x) + r(x)$ for some $q(x), r(x) \in F[x]$ where $\deg r < \deg f$; since $r(x) = g(x) - f(x)q(x)$, the only way this makes sense at all is if r is the zero polynomial, i.e. $g(x)$ is divisible by $f(x)$, which implies that $I = (f(x))$. $\square$

Problem 2. Let R be a PID.

- (a) Show that any two elements $a, b \in R$ have a least common multiple $\text{lcm}(a, b)$, which is unique up to multiplication by units.
- (b) Show that $\gcd(a, b) \text{lcm}(a, b)$ equals ab , up to a unit.

Solution. (a) Let m be a generator of the intersection $(a) \cap (b)$. Then, if M is a common multiple of a and b , by definition $M \in (a)$ and $M \in (b)$, so $M \in (a) \cap (b) = (m)$, i.e. m divides any common multiple M of a and b , making it a least common multiple of a and b .

(b) If g is a greatest common divisor of a and b , then write $a = ga'$, $b = gb'$. Then, $g = as + bt$ for some $s, t \in R$ since $(g) = (a) + (b)$, so $g = g(a's + b't)$, implying that $a's + b't = 1$ because we can cancel the g s due to R being an integral domain. Thus $(a') + (b') = (1)$, so $(a') \cap (b') = (a')(b') = (a'b')$, meaning $\text{lcm}(a', b') = a'b'$.

Now, $(g)((a') \cap (b')) \subseteq (a) \cap (b)$ because if $a'u = b'v \in (a') \cap (b')$ for some $u, v \in R$, then $au = ga'u = gb'v = bv$ is in $(a) \cap (b)$. The reverse inclusion holds because if $au = bv \in (a) \cap (b)$ then $ga'u = gb'v$ implies $a'u = b'v$ due to being in an integral domain, so that $a'u = b'v \in (a') \cap (b')$ and $ga'u = gb'v \in (g)((a') \cap (b'))$.

Therefore, $\text{lcm}(a, b) \gcd(a, b) = g \text{lcm}(ga', gb') = g^2 \text{lcm}(a', b') = ga' \cdot gb' = ab$.

(All gcd and lcm computations in this problem are up to a unit.) $\square$

Problem 3. Let R be an integral domain in which every prime ideal is principal.

- (a) Show that if R is not a PID, then there is an ideal I that is not principal, and is maximal with respect to this property.
- (b) Since I cannot be prime, there are elements $a, b \in R$ with $a, b \notin I$ and $ab \in I$. Show that $(a) + I = (c)$ for some $c \in R$.
- (c) Show that the ideal $J = \{r \in R : rc \in I\}$ is principal.
- (d) Conclude that I itself must be principal, which is a contradiction.

Solution. (a) Let $\{I_\alpha\}_{\alpha \in A}$ be a chain of non-principal ideals of R ; the union $I = \bigcup_{\alpha \in A} I_\alpha$ is also an ideal, and it is non-principal since if $I = (a)$ for some $a \in R$ then $a \in I_\alpha$ for some $\alpha \in A$, so $(a) \subseteq I_\alpha \subseteq I = (a)$, hence I_α must be principal, which it isn't. Therefore, by Zorn's lemma, the collection of non-principal ideals of R has a maximal element with respect to inclusion.

(b) Since $a \notin I$, $(a) + I$ strictly contains I , so by the maximality of I among the non-principal ideals, $(a) + I$ cannot be non-principal, i.e. is principal.

(c) J contains I because if $x \in I$ then $xc \in I$, since ideals absorb multiplication; additionally, J contains b because $bc \in b((a) + I) = (ab) + bI \subseteq I$, using the fact that $ab \in I$, so J strictly contains I , hence is principal.

(d) Let $J = (d)$ for some $d \in R$. If $u \in I \subseteq J$ then $u = cy$ for some $y = dz \in J$, so $u \in (cd)$, hence $I \subseteq (cd)$. On the other hand $d \in J$ means $cd \in I$ so $(cd) \subseteq I$, so $I = (cd)$ is principal. $\square$

Problem 4. Exhibit all the ideals in $F[x]/(p(x))$, where F is a field and $p(x)$ is a polynomial in $F[x]$.

Solution. Ideals in $F[x]/(p(x))$ are in correspondence with ideals of $F[x]$ containing $p(x)$, which are exactly the ideals of the form $(q(x))$ where $q(x)$ is a polynomial dividing $p(x)$, with the correspondence given by sending $(q(x))$ to its image under the projection map $\phi: F[x] \rightarrow F[x]/(p(x))$; this image is just the ideal generated by $\phi(q(x))$. One collection of representatives of the elements of this ideal is the polynomials that divide $p(x)$ and are divisible by $q(x)$. $\square$

Problem 5. Let $p \in \mathbb{Z}$ be a prime with $p \equiv 3 \pmod{4}$. Prove that the quotient ring $\mathbb{Z}[i]/(p)$ is a field with p^2 elements.

Solution. Suppose $p \mid ab$ for $a, b \in \mathbb{Z}[i]$. Then $p^2 = N(p) \mid N(ab) = N(a)N(b)$ where $N(x + yi) = x^2 + y^2$ for $x, y \in \mathbb{Z}$. If p divides $N(x + yi) = x^2 + y^2$ for some $x, y \in \mathbb{Z}$, then either x and y are both multiples of p , or

$$(xy^{-1})^2 \equiv -1 \pmod{p},$$

i.e. -1 is a quadratic residue modulo p . However, the latter is impossible because $p \equiv 3 \pmod{4}$ — to see this, if g is a generator of the cyclic group $(\mathbb{Z}/p\mathbb{Z})^\times$ then $g^{(p-1)/2} = -1$ and $(p-1)/2$ is odd, while all nonzero quadratic residues are expressible as g^k for some even $k \in \{1, \dots, p-1\}$; this means $p \mid x + yi$ is equivalent to p dividing both x and y . Since $p^2 \mid N(a)N(b)$, p divides at least one of $N(a)$ and $N(b)$, so p divides whichever of a and b has norm divisible by p ; thus p is prime in $\mathbb{Z}[i]$.

Now, $\mathbb{Z}[i]/(p) = \{x + yi : x, y \in \{0, \dots, p-1\}\}$ clearly has p^2 elements (noting that these elements all represent different cosets of (p) because the difference between any two distinct ones has norm indivisible by p) and is an integral domain since (p) is a prime ideal, as it is generated by a prime element. Finite integral domains are fields, so $\mathbb{Z}[i]/(p)$ is a field of order p^2 . $\square$

Problem 6. Show that (x, y) is not a principal ideal in $\mathbb{Q}[x, y]$.

Solution. Since $\mathbb{Q}$ is a field, $\mathbb{Q}[x, y]$ is a UFD. x and y are irreducible in this ring because if $x = pq$ for $p, q \in \mathbb{Q}[x, y]$, then one of p and q must be a constant polynomial by degree considerations, and similarly for y . Therefore any common divisor of both x and y is a unit, so if (x, y) is principal then it must equal (1) ; but this is obviously false, since (x, y) doesn't contain any constant polynomials. $\square$

Problem 7. Suppose that $f(x)$ and $g(x)$ are two polynomials with rational coefficients whose product $f(x)g(x)$ has integer coefficients. Show that the product of any coefficient of $f(x)$ with any coefficient of $g(x)$ is an integer.

Solution. Let q_f and q_g be the unique rational numbers such that $q_f f(x)$ and $q_g g(x)$ are both primitive. Then $q_f f(x) \cdot q_g g(x)$ is primitive, so $\frac{1}{q_f q_g}$ is equal to the content of $f(x)g(x)$, which is an integer. If $q_f f(x) = a_n x^n + \dots + a_1 x + a_0$ and $q_g g(x) = b_m x^m + \dots + b_1 x + b_0$ for $a_i, b_j \in \mathbb{Z}$, then the coefficients of f are a_i/q_f and the coefficients of g are b_j/q_g , so the product of some two is

$$\frac{a_i b_j}{q_f q_g} = a_i b_j \cdot \text{cont}(f(x)g(x)) \in \mathbb{Z}. \quad \square$$

Problem 8. Let F be a field. A subring $R \subseteq F$ is called a **valuation ring** if, for every nonzero $x \in F$, at least one of x and x^{-1} belongs to R .

(a) Show that R has a unique maximal ideal.

(b) Show that the ideals of R are totally ordered under inclusion.

Solution. It suffices to show (b), since a totally ordered set has at most one maximal element. Let I and J be two ideals of R that are not comparable under inclusion, so $I \not\subseteq J$ and $J \not\subseteq I$, and let $a \in I \setminus J$, $b \in J \setminus I$ (note $a, b \neq 0$). Then one of $ab^{-1}, a^{-1}b \in F$ is in R , so either $ab^{-1} \cdot b = a \in J$ or $ba^{-1} \cdot a = b \in I$, contradiction; hence the collection of ideals of R is totally ordered. $\square$